Automated scoliosis classification from AI-enabled spine contouring

Type: Poster Presentation
Author(s)/Presenter(s): Lahoti Y, Yu J, Cho SK, Kim JS
First/Senior Author: yes
Event/Meeting: Congress of Neurological Surgeons Annual Meeting
Address: Los Angeles, California, United States
Month/Year: 10/2025

Other Publications

Optic nerve blood flow measurement

Type: Peer-Reviewed Book Chapter
Chapter Author(s): Harris A, Potash S, Verticchio Vercellin A, Lahoti Y, Siesky B
First/Senior Author: no
Book Title: Ophthalmology. 7th Edition
Editor(s): Yanoff M, Duker J
Year: 2026
Publisher(s): Elsevier Publications (Edinburgh)
Publication Status: In Press

HONORS AND INTERESTS

Honors, awards, achievements, research activities, outside interest and hobbies

Honors, Awards, & Achievements

Medical School

- **AI-Ready Mount Sinai - Departmental Champion Scholarship ($10K)** | Scientific Computing and Data (2026)
    - *Support work to expand AI-enabled research within ophthalmology, fulfill departmental data requests, and develop pipelines connecting multimodal ophthalmic imaging with clinical data across health-system repositories.*
- **AI Panelist** - 6th Prostate Cancer Symposium & World Congress of Urologic Oncology (2024)
- **2nd Place Pitch Challenge ($10K) - SpineSight** | Mount Sinai Innovation Partners (2023)
- **NSF National I-Corps Grant ($50K) - SpineSight**| NSF Entrepreneurial Training Program (2023)
    - *Selected for intensive customer discovery and commercial validation of an AI platform for cervical myelopathy with 150+ interviews with patients and clinicians*
- **NSF Regional I-Corps Grant ($5K) - SpineSight** | NSF Entrepreneurial Training Program (2023)
- **Medical Student Training in Aging Research (MSTAR) Scholarship ($6K)**(2023)
- **Targeted Healthcare Innovation Fellowship ($5K)** | Mount Sinai Biodesign - Cohort 3 (2023)

Undergraduate

- **OtoAI: Technology and Innovation Prize** | University of Pennsylvania School of Engineering (2021)

- *Selected out of 100+ teams for best use of technology and innovation; product pitch and demonstration to a panel of 60 engineering industry leaders*

Research Activities

New York Eye And Ear of Mount Sinai (NYEE) | New York, NY (2025 - Present)

Clinical Research Scholar (Machine Learning and AI) - Dr. Alon Harris Lab

Data Standardization and Infrastructure:

- Co-developed a 10-TB multimodal ophthalmic research database integrating fundus, OCT/OCTA, visual fields, and systemic vascular biomarkers across 5+ NIH studies, enabling centralized analysis, data sharing, and quality control
- Co-authored the IRB protocol and created a scalable clinical registry of 1,000+ patients with central retinal artery occlusion across the Mount Sinai Health System, harmonizing longitudinal EHR data with paired ocular imaging
- Contributed to the technical team responsible for sharing health system data with Sight Outcomes Research Collaborative (SOURCE) Consortium

Computer Vision and Deep Learning Model Development:

- Engineered end-to-end software for data standardization, deep-learning architecture design, and model training/fine-tuning to identify fast progressors in glaucoma from multimodal datasets
    - *An EHR-Based Predictive Model to Quantify Multifactorial Risk for Primary Open Angle Glaucoma (Published Abstract)*
- Developed models from synthetic data to automate capillary-level OCTA segmentation and phenotyping in patients with glaucoma and diabetes using the NIH AI-READI dataset.
    - *Capillary level OCTA segmentation and phenotyping in patients with glaucoma and diabetes (Published Abstract)*

Ocular Imaging Device Innovation:

- Performed ocular imaging for patients enrolled in clinical trials at NYEE, utilizing OCT, OCTA, Laser Doppler Holography, OcuMet, TearFilm Imager in healthy and glaucoma patients at the optic nerve head and macula.
- Coordinated $100K budget across 10 product vendors and 100+ components for laser doppler holography; collaborated on optical system assembly, device maintenance, and biomarker development/validation

Grant Writing and Scientific Proposal Development:

- Co-authored grant proposals, developing the machine-learning methods, AI-ready data architecture, and model-validation strategies for research on glaucoma progression and cardio-oculomics.
    - ***National Institutes of Health (NIH) Grant (2026; Under Review)*** | *A Physiology-Enhanced Multimodal Machine Learning Analysis for Predicting Glaucoma Progression in a Large Multicenter Data Consortium*
    - ***BrightFocus Foundation Grant (2026; Submitted)*** | *Integrating Artificial Intelligence and Hemodynamics to Individually Tailor Therapies for Glaucoma*
    - ***NYEE Foundation Grant (2025; Funded)*** | *Connecting Heart and Eye Blood Flow Through a Combination of Artificial Intelligence, Doppler Holography, Echocardiography, and Retrobulbar Imaging*

Dept. Scientific Computing and Data, Icahn SOM at Mount Sinai | New York, NY (01/2026 - Present)

Lead of Medical Student Engagement and Ophthalmology Informatics - Dr. Andrew Deonarine and Dean Patricia Kovatch

AI-Ready Mount Sinai (AIR-MS): cloud-based, multi-modal health data platform

- Drafted proposals for ophthalmic ontology to convert 140+ captured ophthalmic measurements into a common data model compliant across 12.5 million patient records within the health system.
- Developed a formal training program to integrate AIR-MS and clinical informatics competencies into the medical student research curriculum, targeting 100+ users.

- *Hands-on coding workshops geared to using AI agents to write chart-review SQL queries, perform statistical analysis, accelerate scientific research, and generate personalized education material resources.*

Mount Sinai Million Health Discoveries Program: Mitochondrial DNA (mtDNA) Repository

- Co-organized an institutional mtDNA interest group; contributed to study design, data validation, manuscript preparation, and distribution of created mtDNA repository with over 50K+ patients.
    - *Scalable Implementation of Population mtDNA Analysis Across Institutional and National Biobanks (Submitted to Nature Methods)*

Department of Orthopedics, Icahn SOM at Mount Sinai | New York, NY (2022 - 2024)

Lead AI Student Research Director - Dr. Samuel Cho and Dr. Jun Kim

- Led the student research efforts of 10+ AI initiatives and development of 2 software prototypes for spine-surgery clinical workflows
- Established a proprietary longitudinal database of 1,200+ scoliosis cases, radiographic annotations, and clinical outcomes to train surgical prognosis AI models and evaluate against 10K+ active patients across the Sinai health system

SpineSight - Targeted Healthcare Innovation Fellowship, Mount Sinai Biodesign| New York, NY (2022 - 2025)

Co-Founder & Lead Engineer

- Spearheaded the technical development of SpineSight, an AI-driven platform that enhances quantification and diagnosis of spinal cord diseases, automating MRI measurements and reducing patient diagnostic time by 85% (12 months); received national, regional, and institutional innovation funding.

TDK SensEI - End-to-end AutoML Platform for Edge Computing and Machine Intelligence | Pittsburgh, PA (2022)

Machine Learning Engineer - Wearable Sensor Data

- Constructed a TinyML solution for predicting invasive cardiac pressures using non-invasive wearable ECG data, based on over 2000+ hours of patient sensor data, achieving 95% prediction accuracy for central venous pressure in an outpatient facility.

Outside Interests and Hobbies

- **Chai & Coffee Enthusiast:** I enjoy treating my morning pour-over and masala chai routines like a chemistry experiment, and I will always bring hot beverages to share.
- **Bollywood Fusion Dance:** I take local fusion dance classes and am currently choreographing for an upcoming wedding with my college bhangra dance team.
- **Gaming:** I am equally happy getting lost in an open-world game for hours or crowding around the couch with friends for co-op games that quickly become competitive. Favorites include *Spider-Man 2*, *Rocket League*, and *Elden Ring*.
- **Vibe Coding:** Turning rough ideas into functional web apps scratches my data-nerd itch, and I will try to convince you to use the programs I build for fun.

Specialty electives and related activities

Clinical Ophthalmology Electives

Ophthalmology Elective

New York Eye and Ear Infirmary of Mount Sinai | May 2025 (2 weeks)

- Introductory exposure to ophthalmology across subspecialties; participated in clinic and observed multiple surgical procedures

Ophthalmology Elective

New York Eye and Ear Infirmary of Mount Sinai | June 2026 (4 weeks)

- Participated in outpatient clinics, surgical cases, and resident education across multiple ophthalmic subspecialties; presented original research at *Resident Research Day*.

Clinical Ophthalmology Elective

SUNY Downstate Health Sciences University and Kings County Hospital | July 2026 (4 weeks)

- Independently performed initial patient evaluations and presented findings and management plans to residents and faculty; obtained multimodal imaging in resident and subspecialty clinics.

Advanced Ophthalmology Elective

NYU Langone Health and Bellevue Hospital | August 2026 (4 weeks)

- Participated and observed in subspecialty clinics, hospital consults, and surgical care; attended and presented at *NYU Ophthalmology Grand Rounds*.

Ophthalmology Elective: General

Northwell Health and the Zucker School of Medicine at Hofstra/Northwell | September 2026 (4 weeks, scheduled)

- Future elective with planned participation in subspecialty clinics and hospital consults.

Ophthalmology Mentorship and Education

Founding Member | Society for Artificial Intelligence in Vision and Ophthalmology (SAIVO) (2025 - Present)

- Initial 24 members; Contributed to development of educational media at the intersection of artificial intelligence, vision science, and clinical ophthalmology for trainees

Senior Clinician (2025); Teaching Senior (2026) - East Harlem Health Outreach Partnership (EHHOP) at Mount Sinai | New York, NY (2025 - Present)

- Participated in six student-run ophthalmology clinics for uninsured patients in East Harlem community
- Led teaching sessions on slit-lamp examination, refraction, ophthalmic examination skills, and clinical presentation

Volunteer Undergraduate Ophthalmology Research Mentor | Greater New York Area (2026 - Present)

- Mentor 3 high school and undergraduate students biweekly in computer science, mathematics, and physics to guide personalized AI research projects on open-source ophthalmic datasets

Academic Service / Scholarly Review

- **Peer Reviewer,** *Scientific Reports* (2026)
- **Peer Reviewer**, *Artificial Intelligence in Vision and Ophthalmology* (AIVO) (2025 - Present)

Public service and activities

Clinical Service and Community Health

University of Pennsylvania Medical Emergency Response Team | Philadelphia, PA (2018 - 2021)

Emergency Medical Technician-Basic

- Completed 750 hours of volunteer service and responded to more than 60 emergency calls, providing patient care and detailed documentation in high-stress, time-sensitive situations.
- Co-founded Mergineers, an analytics committee that used emergency-response data to identify opportunities to improve campus patient care.

Indira Gandhi Government Medical College and Hospital | Nagpur, India (2020)

COVID-19 Vaccination Survey and Community Education Volunteer

- Organized volunteers to survey 2,500 villagers, digitized written responses, and created accessible infographics explaining vaccination benefits and comparing the cost of vaccination with hospitalization.

Education and Mentorship

Admissions Committee Member, Icahn SOM at Mount Sinai | New York, NY (2025 - 2026)

- Participated in applicant review of 30+ medical school applications as a student member of the admissions committee.

***"Artificial Intelligence in Medicine"* - NEXUS Course Co-Director**, Icahn SOM at Mount Sinai (*Dr. Girish Nadkarni - Faculty Mentor*) | New York, NY (2022 - 2024)

- Designed and directed a new medical school elective course for 50+ first- and second-year students, including 8 interactive coding workshops and a guest-lecture series featuring F500 industry leaders.

AI Lead Course Instructor, InspiritAI | Philadelphia, PA (2021)

- Taught 20 high school students foundational mathematical and computational concepts in artificial intelligence over a 3-month summer course.

Conference Committee Co-founder and Outreach Chair, UPenn South Asian Society | Philadelphia, PA (2019)

- Co-founded and organized the inaugural South Asian Leadership Conference, connecting more than 100 students with South Asian professionals and mentors; the conference subsequently became an annual event.

Neuroscience Instructor, Neuroscience Pipeline Program | Philadelphia, PA (2017 - 2018)

- Developed activity-based curricula and taught neuroscience concepts to underserved high school students in West Philadelphia, encouraging curiosity and continued education in STEM.

Additional Information

Do you currently have a partner participating in the Ophthalmology residency match?	**No**

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Permanent Address: 1-15 57th Avenue, APT 1426, Queens, New York 11101, United States

EDUCATION

07/2022 - 05/2027 Icahn School of Medicine at Mount Sinai

Medical School

Address: Icahn School of Medicine at Mount Sinai- Off. Student Aff.,One Gustave L. Levy Place, Box 1257, New York, New York 10029, United States
AAMC ID: 15085176
Degrees: MD
GPA:
Class Rank:
AOA: Not available at my institution
Gold Humanism: No, I was not selected.

05/2021 - 12/2021 University of Pennsylvania School of Engineering and Applied Science

Graduate education (non-medical)

Address: Philadelphia, Pennsylvania 19104, United States
Degrees: MS
GPA: 3.87
Major: Electrical Engineering

08/2017 - 05/2021 University of Pennsylvania School of Engineering and Applied Science

Undergraduate education

Address: Philadelphia, Pennsylvania 19104, United States
Degrees: BS
Honors: Summa Cum Laude
GPA: 3.86
Major: Biomedical Science

EXAMINATIONS

USMLE ID: 55433429

Step	Date	3 Digit Score	Times taken	Test results
Step 1	06/10/2024		1	Pass
Step 2 CK	05/23/2026	249	1	

EMPLOYMENT

07/2025 - 07/2026 **Clinical Research Scholar**

Medical career related

Employer name: The Barry Family Center for Ophthalmic Artificial Intelligence and Human Health
Address: 310 East 14th Street, Manhattan, New York, New York 10013, United States

01/2026 - 06/2026 **Ophthalmology & AI Consultant**

Medical career related

Employer name: Theta Tech AI
Address: Headquarters Greater New York City, New York, New York n/a, United States

10/2022 - 01/2026 **Academic & Admissions Consultant**

Employer name: Sapient Academy
Address: 930 Roosevelt, Irvine, California 92620, United States

01/2022 - 07/2022 **Machine Learning Engineer**

Employer name: TDK SensEI
Address: 6101 Penn Avenue, Pittsburgh, Pennsylvania 15206, United States